Table 8. Experimental validation evidence

Panel A. System and reported result

Dataset id	Exact chemical or biological system	Observable	Conditions	Reported result	Unit
beef_heart_submitochondrial_superoxide	beef heart submitochondrial superoxide	superoxide proxy rate	dose=90 uM doxorubicin; oxygen=not reported in accessible abstract	control: 1.6; doxorubicin: 69.6; uncertainty SE: 0.2, 2.7	nmol min^-1 mg^-1
beef_heart_submitochondrial_H2O2	beef heart submitochondrial H2O2	hydrogen peroxide (H2O2)	dose=200 uM doxorubicin; oxygen=not reported in accessible abstract	control: undetectable; doxorubicin: 2.2; uncertainty SE: 0.3	nmol min^-1 mg^-1
Ehrlich_microsome_superoxide	200 ug microsomal protein in 1 mL; 150 mM potassium phosphate; 100 uM EDTA; 1 mM NADPH	superoxide proxy rate	dose=135 uM; environment=200 ug microsomal protein in 1 mL; 150 mM potassium phosphate; 100 uM EDTA; 1 mM NADPH; temperature=310 K; pH=7.4; oxygen=not separately reported	control: 0.51; doxorubicin: 14.71; n: 3, 6; uncertainty SE: 0.26, 1.43	nmol min^-1 mg^-1
Ehrlich_mitochondrial_superoxide	100 ug protein/mL; 250 mM sucrose; 20 mM HEPES; 100 uM EDTA; NADH; 4 uM rotenone; 5 min preincubation	superoxide proxy rate	dose=135 uM; environment=100 ug protein/mL; 250 mM sucrose; 20 mM HEPES; 100 uM EDTA; NADH; 4 uM rotenone; 5 min preincubation; temperature=310 K; pH=8.2; oxygen=not separately reported	control: 1.12; doxorubicin: 7.29; n: 7, 10; uncertainty SE: 0.15, 0.61	nmol min^-1 mg^-1
Ehrlich_nuclear_superoxide	200 ug nuclear protein; sucrose/20 mM HEPES/100 uM EDTA; 1 mM NADPH	superoxide proxy rate	dose=135 uM; environment=200 ug nuclear protein; sucrose/20 mM HEPES/100 uM EDTA; 1 mM NADPH; temperature=310 K; pH=7.4; oxygen=not separately reported	control: 0.36; doxorubicin: 3.29; n: 7, 12; uncertainty SE: 0.05, 0.33	nmol min^-1 mg^-1
Ehrlich_H2O2	fraction buffers as separately reported; cells in Chelex-treated PBS	hydrogen peroxide (H2O2)	dose=135 uM fractions; 400 uM cells; environment=fraction buffers as separately reported; cells in Chelex-treated PBS; temperature=310 K; pH=7.4 microsomes;8.2 mitochondria;cell PBS pH not reported; oxygen=cells air-bubbled 30 min	uncertainty SE: 0.48, 0.6, 0.04; values: 3.42, 2.6, 0.64	nmol min^-1 mg^-1; nmol min^-1 per 1e7 cells
PC3_intracellular_H2O2	PC3 human prostate cancer cells	hydrogen peroxide (H2O2)	dose=1 uM doxorubicin; environment=PC3 human prostate cancer cells; temperature=310 K; oxygen=not reported	basal: 13; doxorubicin: 51; time: 30 min; uncertainty: 4, 13	pM

The records distinguish exact chemical systems and direct validation eligibility. Presentation values are rounded. All 9 source rows and all 24 source columns remain in table08_experimental_validation_evidence.csv; 14 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 8. Experimental validation evidence

Panel A. System and reported result (continued)

Dataset id	Exact chemical or biological system	Observable	Conditions	Reported result	Unit
H9c2_MitoSOX	HBSS with Ca/Mg and 1% BSA; 5 uM MitoSOX	relative MitoSOX fluorescence (oxidant proxy)	dose=10/20/50 uM doxorubicin; environment=HBSS with Ca/Mg and 1% BSA; 5 uM MitoSOX; temperature=310 K; oxygen=not reported	uncertainty: 0.04, 0.08, 0.08; values: 1.4, 1.8, 2.8	fold control
quinone_disabled_negative_control	Ehrlich microsomal system	superoxide proxy rate	dose=135 uM both compounds; environment=Ehrlich microsomal system; temperature=310 K; pH=7.4; oxygen=not separately reported	doxorubicin: 14.71; five iminodaunorubicin: 0.45; uncertainty SE: 1.43, 0.15	nmol min^-1 mg^-1

The records distinguish exact chemical systems and direct validation eligibility. Presentation values are rounded. All 9 source rows and all 24 source columns remain in table08_experimental_validation_evidence.csv; 14 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.

Table 8. Experimental validation evidence

Panel B. Method and validation use

Dataset id	Method	Source	Usable for direct validation	Limitations
beef_heart_submitochondrial_superoxide	SOD-inhibitable cytochrome-c reduction	doi:10.1016/S0021-9258(17)35747-2	no	Conditions and normalization differ; do not pool or fit.
beef_heart_submitochondrial_H2O2	catalase-released oxygen	doi:10.1016/S0021-9258(17)35747-2	no	Detection limit and exact conditions not reported in accessible abstract.
Ehrlich_microsome_superoxide	SOD-inhibitable acetylated cytochrome-c reduction	doi:10.1155/2019/9474823	no	Conditions and normalization differ; do not pool or fit.
Ehrlich_mitochondrial_superoxide	SOD-inhibitable acetylated cytochrome-c reduction	doi:10.1155/2019/9474823	no	Conditions and normalization differ; do not pool or fit.
Ehrlich_nuclear_superoxide	SOD-inhibitable acetylated cytochrome-c reduction	doi:10.1155/2019/9474823	no	Conditions and normalization differ; do not pool or fit.
Ehrlich_H2O2	catalase-released oxygen; linear 10-30 min interval	doi:10.1155/2019/9474823	no	Conditions and normalization differ; do not pool or fit.
PC3_intracellular_H2O2	catalase-aminotriazole kinetic assay	doi:10.1016/j.abb.2005.06.015	no	Apparent intracellular H2O2; medium Amplex Red did not increase and catalase overexpression did not increase resistance.
H9c2_MitoSOX	1 h fluorescence	doi:10.1016/j.bbrc.2007.04.106	no	Sensitivity only; autofluorescence and nonspecific probe products prohibit absolute calibration.
quinone_disabled_negative_control	acetylated cytochrome-c reduction	doi:10.1155/2019/9474823	no	Supports quinone dependence but not D/Q selectivity.

The records distinguish exact chemical systems and direct validation eligibility. Presentation values are rounded. All 9 source rows and all 24 source columns remain in table08_experimental_validation_evidence.csv; 14 provenance/support columns are source-only. 'Not available' is distinct from numeric zero.